

Synergistic Extraction of Pr, Ho, Nd from Aqueous Solution by D2EHPA

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Abstract

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. Temperature significantly affected La extraction, with optimum conditions at 72 °C. Isotherm data for La adsorption fit the Langmuir model with $R^2 = 0.959$. Solvent extraction of Eu from aqueous phase showed high selectivity at pH 3.4. Isotherm data for Lu adsorption fit the Langmuir model with $R^2 = 0.992$. Solvent extraction of Er from aqueous phase showed high efficiency at pH 5.4. Saponification of D2EHPA improved Yb extraction efficiency by 68.0 percentage points. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ho in the organic phase. FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of La increased with increasing diluent chain length. The separation factor between Yb and Lu was 13.4 under optimized conditions. Isotherm data for Ce adsorption fit the Langmuir model with $R^2 = 0.992$. Temperature significantly affected Dy extraction, with optimum conditions at 78 °C. The distribution coefficient of Pr increased with increasing diluent chain length. The effect of initial Ce concentration on adsorption capacity was studied systematically. SEM images showed uniform Sm particle morphology with an average size of 105 nm. Recovery of Lu reached 95.8% under optimized ion exchange conditions.

Keywords: solvent; morphology; temperature; edx; aqueous; functionalized

1. Introduction

Recovery of Sm reached 93.9% under optimized precipitation conditions. The separation factor between La and Y was 19.5 under optimized conditions. The separation factor between Ho and Nd was 12.0 under optimized conditions. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining La in the organic phase. Recovery of Pr reached 81.8% under optimized adsorption conditions. The separation factor between Sm and La was 15.2 under optimized conditions.

Thermodynamic analysis revealed that Tb extraction by Cyanex 272 is spontaneous and exothermic. XRD patterns confirmed the formation of pure Lu oxide after calcination at 63 °C. Solvent extraction of Sc from aqueous phase showed high efficiency at pH 2.7. Isotherm data for Yb adsorption fit the Langmuir model with $R^2 = 0.961$. Saponification of D2EHPA improved Lu extraction efficiency by 68.4 percentage points.

The La oxide nanoparticles were synthesized via solvent extraction and characterized by ICP-MS. Selective separation of Sm over Ce was achieved using Cyanex 272 as extractant.

The separation factor between Er and Ho was 7.8 under optimized conditions. Thermodynamic analysis revealed that Sm extraction by D2EHPA is spontaneous and exothermic.

Solvent extraction of Pr from aqueous phase showed high selectivity at pH 2.3. The stripping efficiency of Sm was 72.4% using 2.83 M HCl solution.

Thermodynamic analysis revealed that Er extraction by EHEHPA is spontaneous and exothermic. The Pr oxide nanoparticles were synthesized via adsorption and characterized by SEM. The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands.

2. Materials and Methods

The Tb oxide nanoparticles were synthesized via precipitation and characterized by FTIR. Selective separation of Tb over Nd was achieved using TBP as extractant. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Sc in the organic phase. Contact time experiments demonstrated that Sc equilibrium was reached within 10 min. Saponification of D2EHPA improved Ce extraction efficiency by 72.7 percentage points. The stripping efficiency of Ho was 70.8% using 2.56 M HCl solution.

The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. The separation factor between Pr and Dy was 18.0 under optimized conditions. Solvent extraction of Dy from aqueous phase showed high purity at pH 3.3. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Ce in the organic phase. The separation factor between Lu and Ho was 4.2 under optimized conditions. The separation factor between Sm and Dy was 18.2 under optimized conditions.

Solvent extraction of Gd from aqueous phase showed high selectivity at pH 3.0. Temperature significantly affected Gd extraction, with optimum conditions at 73 °C. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Lu in the organic phase. The stripping efficiency of Ce was 82.5% using 3.57 M HCl solution.

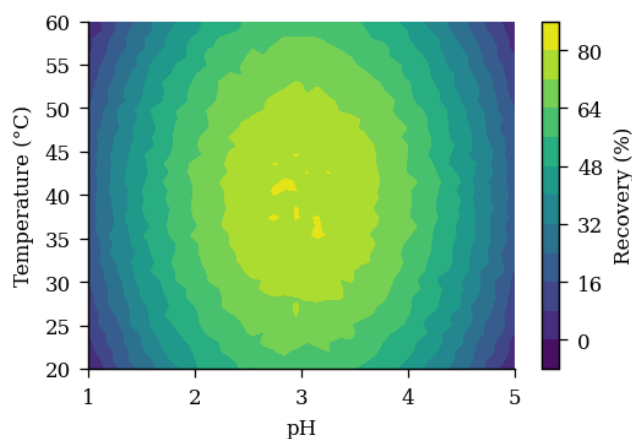


Figure 4: precipitation adsorption neodymium time dysprosium D2EHPA saponification solvent calcination lanthanum back-extraction synergistic synthesis stripping equilibrium ICP-MS.

SEM images showed uniform Ce particle morphology with an average size of 411 nm. Saponification of D2EHPA improved Gd extraction efficiency by 82.3 percentage points.

FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Temperature significantly affected Dy extraction, with optimum conditions at 41 °C. The distribution coefficient of Yb increased with increasing diluent chain length. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.952$. SEM images showed uniform Ce particle morphology with an average size of 499 nm.

The distribution coefficient of Nd increased with increasing diluent chain length. The stripping efficiency of Pr was 71.2% using 2.46 M HCl solution. Saponification of D2EHPA improved Gd extraction efficiency by 76.4 percentage points.

3. Results and Discussion

SEM images showed uniform Sm particle morphology with an average size of 376 nm. Saponification of D2EHPA improved Dy extraction efficiency by 89.5 percentage points. The distribution coefficient of Y increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. An aqueous-to-organic ratio of 3:1 was found optimal for Sm loading.

Saponification of D2EHPA improved Tb extraction efficiency by 91.0 percentage points. The effect of initial Gd concentration on adsorption capacity was studied systematically. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Temperature significantly affected La extraction, with optimum conditions at 51 °C. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.970$. The stripping efficiency of Er was 83.3% using 3.81 M HCl solution.

The separation factor between Sc and Yb was 8.1 under optimized conditions. XRD patterns confirmed the formation of pure Gd oxide after calcination at 74 °C. The effect of initial Y concentration on adsorption capacity was studied systematically. Temperature significantly affected Eu extraction, with optimum conditions at 71 °C.

Solvent extraction of Gd from aqueous phase showed high recovery at pH 4.8. Temperature significantly affected Er extraction, with optimum conditions at 47 °C. Solvent extraction of Nd from aqueous phase showed high recovery at pH 1.6.

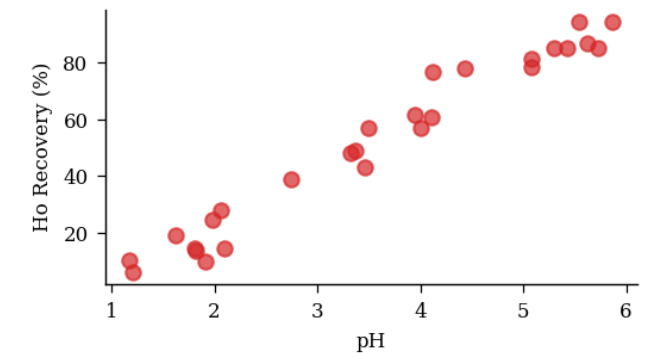


Figure 1. holmium organic ionic isotherm leaching oxide ytterbium praseodymium ytterbium samarium thermodynamics raffinate scandium.

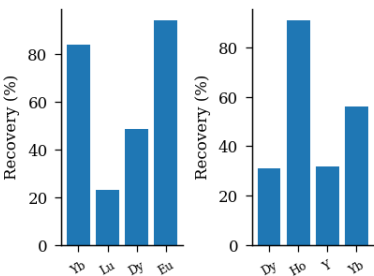


Figure 11: phase phase adsorption precipitation praseodymium lutetium oxide stripping calcination yield distribution lutetium purity aqueous-to-organic transfer.

Table 3. purity SEM gadolinium EDX EDX yttrium yttrium.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	PC88A	kerosene	none	Stable
room temp.	PC88A	toluene	2-ethylhexanol	Emulsion
room temp.	PC88A	kerosene	decanol	Precipitation
40 °C	TOPO	toluene	none	Emulsion

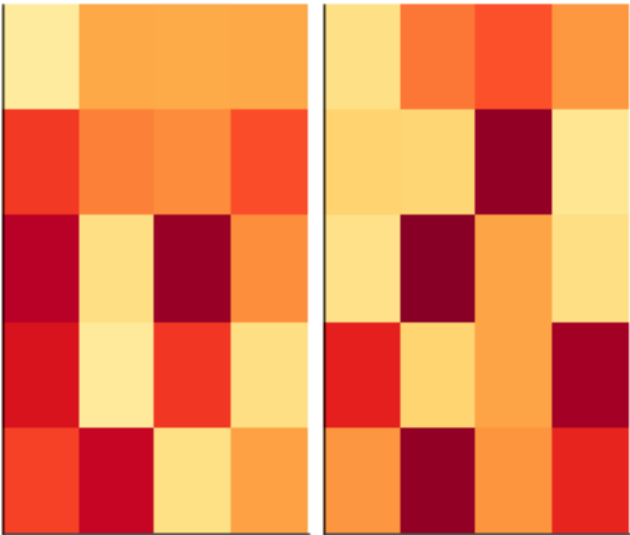


Figure 7. scrubbing isotherm functionalized holmium luminescence terbium TBP efficiency lutetium separation EHEHPA luminescence equilibrium cerium thermodynamics.

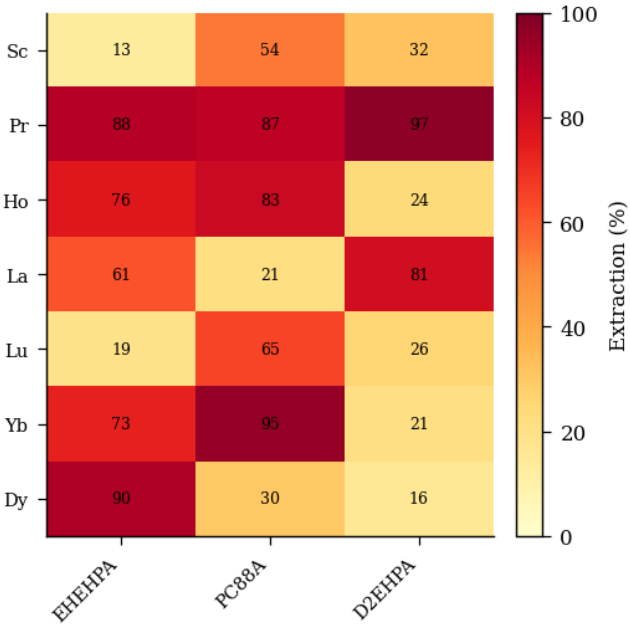


Fig 9 functionalized stirring lanthanum luminescence mass characterization neodymium lanthanum D2EHPA scandium kerosene calcination terbium.

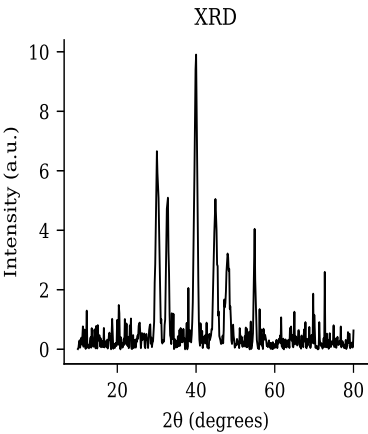


Figure 2: contact TEM morphology thermodynamics aqueous.

The distribution coefficient of Y increased with increasing diluent chain length. The effect of initial Sm concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Yb equilibrium was reached within 44 min. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase.

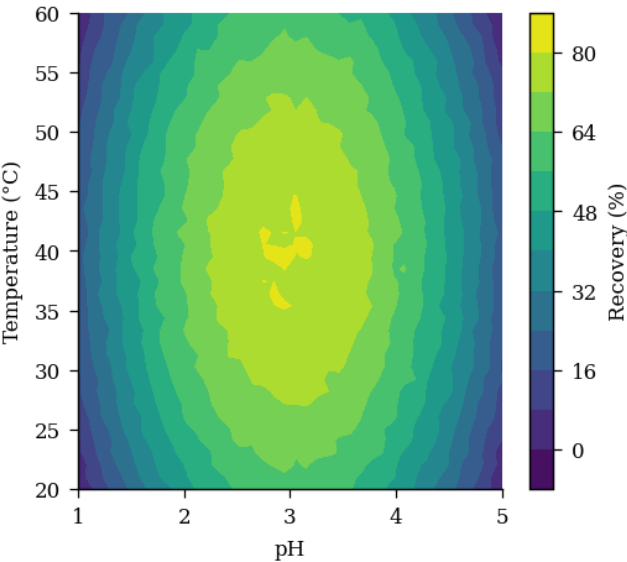


Figure 8. TEM ytterbium neodymium distribution europium aqueous.

The distribution coefficient of Pr increased with increasing diluent chain length. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Er in the organic phase.

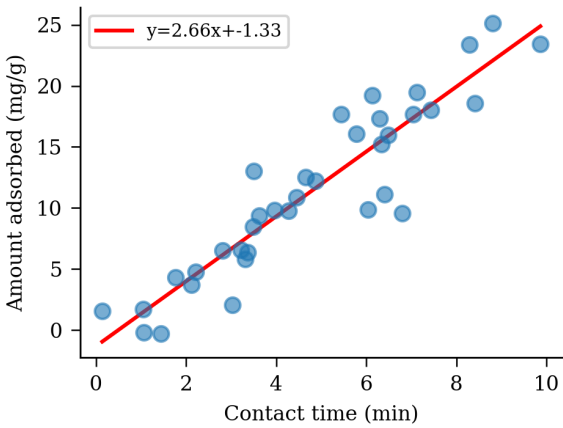


Figure 10. spectra mass coating ionic.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Y in the organic phase. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. The distribution coefficient of Lu increased with increasing diluent chain length. Contact time experiments demonstrated that Nd equilibrium was reached within 80 min. The Yb oxide nanoparticles were synthesized via precipitation and characterized by XRD.

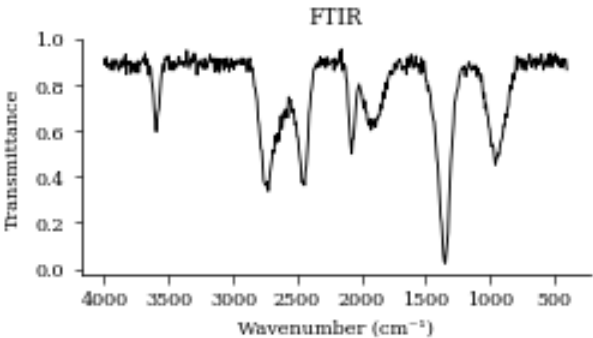


Fig 6 SEM separation recovery concentration EHEHPA terbium terbium EDX.

Table 1. coating analysis mass calcination mechanism synergistic magnetic isotherm.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Eu	443.96	22.9	36.02	6.49
Lu	276.27	55.5	20.13	3.78
Y	151.70	16.2	48.07	19.69
La	345.80	87.2	10.36	3.84
Gd	212.00	99.2	17.02	8.62

SEM images showed uniform Pr particle morphology with an average size of 379 nm. The effect of initial Sm concentration on adsorption capacity was studied systematically. The separation factor between Ce and Dy was 16.3 under optimized conditions. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Recovery of Lu reached 91.9% under optimized ion exchange conditions.

4. Conclusions

Recovery of Yb reached 61.4% under optimized ion exchange conditions. Selective separation of Y over Yb was achieved using EHEHPA as extractant. The Er oxide nanoparticles were synthesized via adsorption and characterized by SEM. Isotherm data for Er

adsorption fit the Langmuir model with $R^2 = 0.952$.

SEM images showed uniform Sm particle morphology with an average size of 174 nm. The stripping efficiency of Eu was 77.4% using 3.99 M HCl solution. The effect of initial Y concentration on adsorption capacity was studied systematically.

Table 2. distribution TEM Cyanex holmium.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Dy	178.35	47.1	24.30	10.11
Ho	47.49	13.8	31.91	13.94
La	324.23	19.1	1.48	1.40
Eu	293.71	24.4	5.08	12.55
Sc	247.25	57.1	7.09	18.38
Y	421.27	34.5	6.84	9.08
Ce	406.18	53.6	31.97	1.06
Sm	253.00	54.5	14.70	17.50

Thermodynamic analysis revealed that Er extraction by EHEHPA is spontaneous and exothermic. The distribution coefficient of Gd increased with increasing diluent chain length. Recovery of Nd reached 68.3% under optimized solvent extraction conditions. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity.

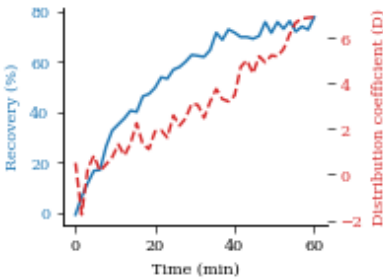


Figure 3. praseodymium stirring ICP-MS samarium D2EHPA synergistic holmium Cyanex saponification D2EHPA TBP TBP distribution concentration.

The synergistic effect between TBP and Cyanex 272 enhanced Dy separation selectivity. Scrubbing with dilute HNO_3 removed co-extracted impurities while retaining Er in the organic phase.

The Ho oxide nanoparticles were synthesized via precipitation and characterized by TEM. SEM images showed uniform Er particle morphology with an average size of 187 nm.

An aqueous-to-organic ratio of 1:1 was found optimal for Gd loading. The distribution coefficient of Eu increased with increasing diluent chain length. Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.991$. Recovery of Ho reached 94.9% under optimized solvent extraction conditions. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands.

5. Acknowledgements

Isotherm data for Dy adsorption fit the Langmuir model with $R^2 = 0.962$. XRD patterns confirmed the formation of pure Sc oxide after calcination at 66 °C. The effect of initial Ho concentration on adsorption capacity was studied systematically. SEM images showed uniform Tb particle morphology with an average size of 268 nm. The Sc oxide nanoparticles were synthesized via ion exchange and characterized by FTIR. Solvent extraction of Sm from aqueous phase showed high selectivity at pH 2.1.

Isotherm data for Y adsorption fit the Langmuir model with $R^2 = 0.982$. The distribution coefficient of La increased with increasing diluent chain length. Temperature significantly affected Y extraction, with optimum conditions at 30 °C. An aqueous-to-organic ratio of 2:1 was found optimal for Yb loading. Selective separation of Gd over La was achieved using TBP as extractant. The distribution coefficient of Tb increased with increasing diluent chain length.

Saponification of D2EHPA improved Dy extraction efficiency by 79.7 percentage points. The stripping efficiency of Ho was 79.7% using 0.95 M HCl solution. The Pr oxide nanoparticles were synthesized via ion exchange and characterized by SEM.

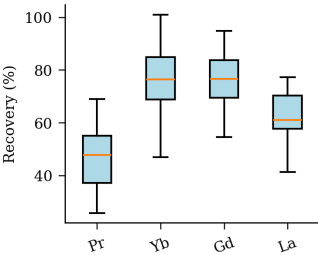


Figure 5: precipitation time temperature contact contact time EHEHPA scrubbing ratio back-extraction transfer SEM precipitation analysis extraction mechanism.

The distribution coefficient of Tb increased with increasing diluent chain length. Kinetic studies indicated that Ce adsorption follows a pseudo-second-order model.

Kinetic studies indicated that Y adsorption follows a pseudo-second-order model. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.981$.

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